

DATA-DRIVEN ANALYSIS OF DIGITAL BANKING SERVICES IN INDIA

Professional Project Report

A Data Analytics Project Using Python and Power BI

Prepared Using



Python



Pandas



NumPy



Matplotlib



Seaborn



Google Colab



Power BI

Project Description

Banking Infrastructure • Digital Payments
Interactive Power BI Dashboard

Project Overview and Description

Digital banking has transformed the financial sector in India by making banking services faster, more convenient, and easily accessible. The increasing use of ATMs, POS terminals, debit cards, credit cards, UPI, internet banking, and mobile banking has significantly improved the way customers perform financial transactions. As digital banking continues to expand, banks generate large volumes of data that can be analyzed to understand customer behavior, transaction patterns, and banking performance.

This project presents a comprehensive analysis of India's digital banking ecosystem using **Python** for data preprocessing, exploratory data analysis (EDA), and visualization, and **Power BI** for interactive dashboard development. The analysis focuses on banking infrastructure, digital payment systems, and online transaction trends to identify meaningful patterns and generate business insights that support data-driven decision-making.

Business Problem

Banks generate a large amount of digital banking data every day through ATMs, POS terminals, debit cards, credit cards, and online transactions. Without proper analysis, it becomes difficult to identify banking trends, compare the performance of different banks, and make informed business decisions. This project analyzes banking data to understand digital banking adoption, evaluate banking infrastructure, and provide valuable insights that can help improve banking services and strategic planning.

Project Objectives

- Analyze banking infrastructure across major Indian banks.
- Compare ATM networks and POS terminal deployment.
- Study debit card and credit card usage.
- Analyze online transaction trends over time.
- Compare the performance of different bank categories.

- Develop interactive Power BI dashboards to communicate business insights.

Dataset & Data Preparation

The analysis uses a cleaned banking dataset containing information about bank names, bank categories, ATM networks, POS terminals, debit cards credit cards, online transaction values, UPI QR deployment, and ATM withdrawal transactions. Before visualization, the dataset was prepared in Python to improve quality and consistency.

Dataset Details

- Source: India Data Portal
- Rows: 2,671
- Columns: 30
- Time Period: 2022–2025
- File Format: CSV

Step	Data Preparation Activity	Purpose
1	Imported the dataset	Load banking data into Python
2	Handled missing values	Improve data quality
3	Removed duplicate records	Ensure data accuracy
4	Corrected data types	Enable accurate calculations
5	Performed exploratory data analysis (EDA)	Understand patterns and relationships
6	Prepared cleaned dataset	Used for Power BI and dashboard development

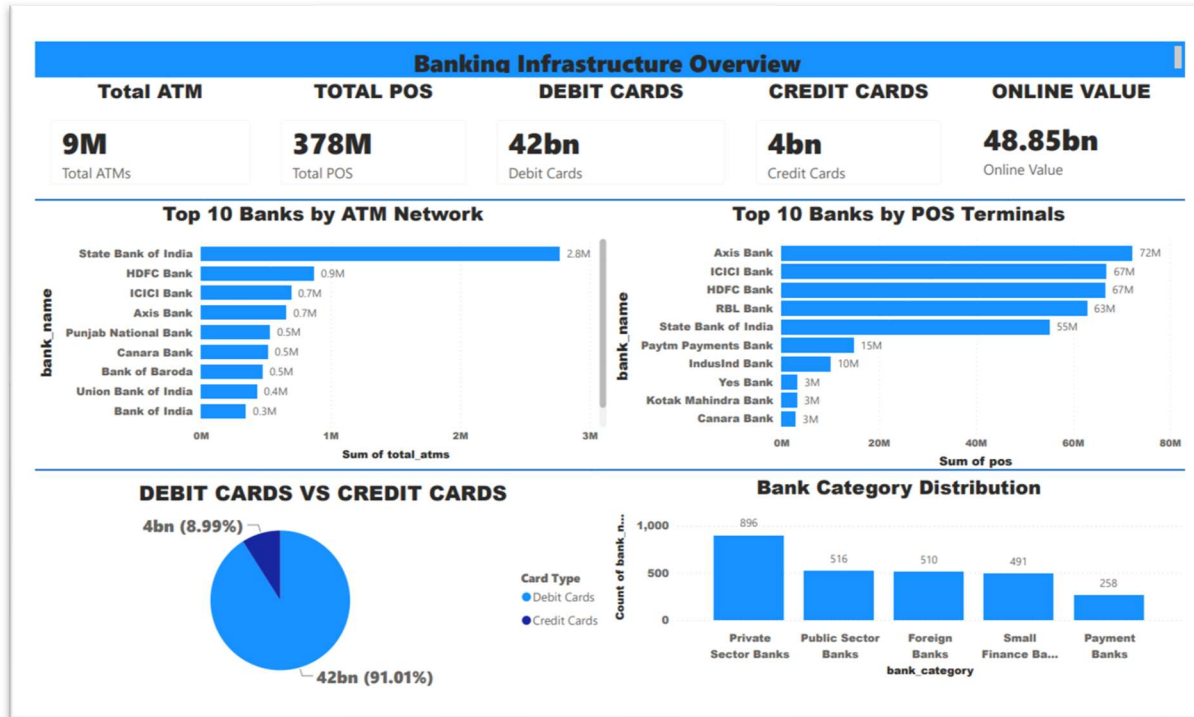
Tools & Technologies

Tool	Purpose
Python	Data preprocessing and analysis
Pandas	Data manipulation
NumPy	Numerical computation
Matplotlib	Visualization
Seaborn	Statistical visualization
Google Colab	Python development environment
Power BI	Interactive dashboard creation

Outcome of Data Preparation

After completing the preprocessing steps, the dataset became clean, consistent, and ready for analysis. This enabled the creation of accurate visualizations and interactive dashboards that provide meaningful insights into India's banking infrastructure, digital payment systems, and customer transaction behavior.

Dashboard 1 - Banking Infrastructure Overview



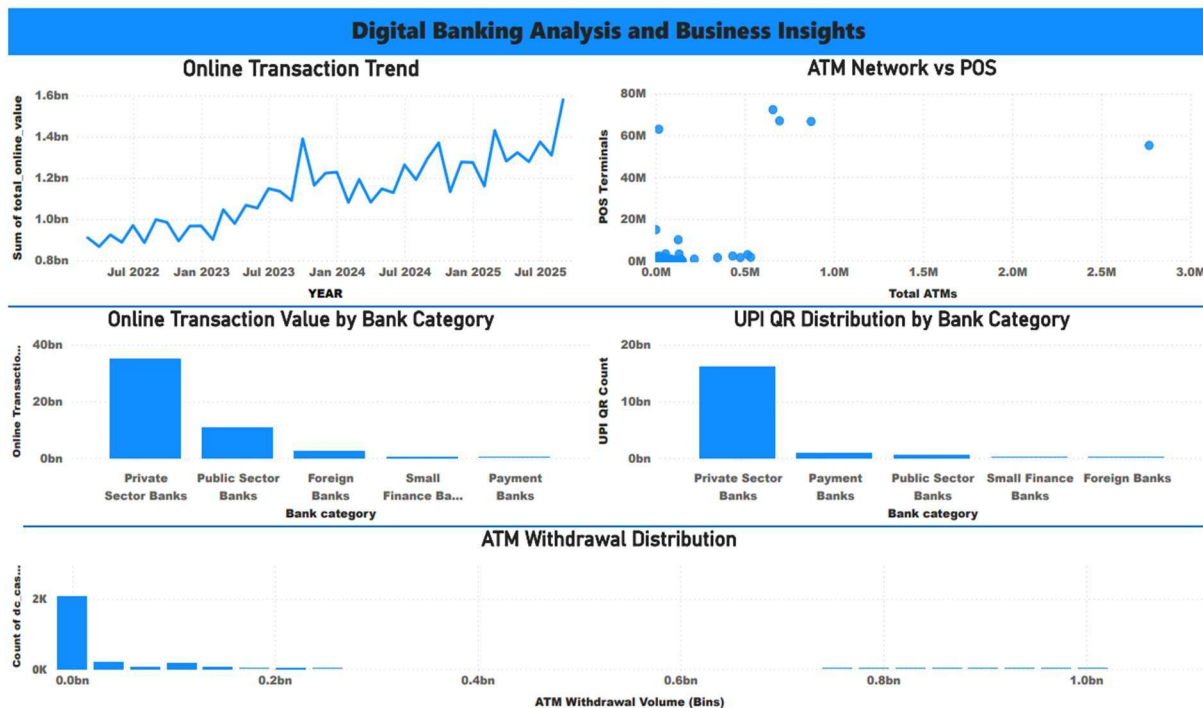
Dashboard Explanation

The first dashboard provides an overview of India's banking infrastructure. It displays key performance indicators (KPIs), including the total number of ATMs, POS terminals, debit cards, credit cards, and online transaction value. It also compares the top 10 banks by ATM network and POS terminals, shows the distribution of debit and credit cards, and presents the distribution of different bank categories. This dashboard helps us understand the overall banking infrastructure and compare the performance of major banks in India.

Key Insights

- The dashboard shows 9 million ATMs, 378 million POS terminals, 42 billion debit cards, 4 billion credit cards, and 48.85 billion in online transaction value.
- State Bank of India has the largest ATM network, while Axis Bank has the highest number of POS terminals.
- Debit cards make up 91.01% of all cards, showing they are used much more than credit cards.
- Private Sector Banks have the highest number of records among all bank categories, indicating a strong presence in the banking sector.

Dashboard 2 – Digital Banking Analysis & Business Insights | Power BI



Dashboard Explanation

The second dashboard focuses on analyzing the growth of digital banking services in India. It includes charts showing online transaction trends, the relationship between ATM networks and POS terminals, online transaction values by bank category, UPI QR distribution, and ATM withdrawal distribution. These visualizations help us understand how digital banking is growing, compare the performance of different bank categories, and identify customer transaction patterns. Overall, the dashboard provides useful insights that support better business decisions.

Key Insights

- Online transaction values have increased steadily, showing the growth of digital banking services.
- Banks with more ATMs generally have more POS terminals, indicating strong banking infrastructure.
- Private Sector Banks have the highest online transaction values and the largest UPI QR deployment, reflecting strong digital payment adoption.
- Most ATM withdrawals are concentrated in lower-value ranges, while

high-value withdrawals occur less frequently.

Project Outcome

This project successfully transformed raw banking data into meaningful insights using Python and Power BI. The dashboards enable users to compare banking performance, monitor digital banking growth, analyze customer transaction patterns, and support data-driven decision-making.

Business Recommendations

- Expand POS terminals and UPI QR infrastructure across rural and semi-urban areas.
- Encourage digital banking adoption through customer awareness programs.
- Promote responsible credit card usage with attractive reward programs.
- Strengthen cybersecurity to protect increasing online transactions.
- Continue investing in digital banking technologies to improve customer experience.

Final Conclusion & Business Impact

This project demonstrates how Python and Power BI can be effectively combined to analyze India's digital banking ecosystem and transform raw banking data into meaningful business insights. Through data cleaning, exploratory data analysis, and interactive dashboard development, the project provides a clear understanding of banking infrastructure, digital payment adoption, and transaction trends.

The analysis shows that Private Sector Banks lead in digital banking services, including POS terminals, online transaction values, and UPI QR deployment, while Public Sector Banks maintain extensive ATM networks, ensuring widespread banking accessibility. The continuous growth in online transactions reflects the increasing adoption of digital banking services across India. The interactive dashboards help decision-makers monitor banking performance, identify trends, and support strategic planning.

Business Impact

- Successfully analyzed India's banking infrastructure and digital payment ecosystem.
- Developed interactive Power BI dashboards for business reporting.
- Identified trends in ATM networks, POS terminals, and online transactions.
- Generated meaningful insights to support data-driven decision-making.
- Demonstrated end-to-end data analytics skills using Python and Power BI.

Future Scope

- Integrate real-time banking data for live analysis.
- Apply machine learning models to predict transaction growth.
- Develop fraud detection and risk analysis models.
- Publish dashboards to Power BI Service for real-time monitoring.
- Expand the analysis with customer-level transaction data.

Project Limitations

- Analysis is based on historical data
- Customer-level information was not available.
- External economic factors were not included.
- Results depend on the quality of the available dataset.